

The pyflightstream guides, guide 5 of 8, numbered 0 to 7

Guide 5: Post-processing definitions

What every product, reduction and window means

Written for pyflightstream 0.31.0.

Source: the package page “Post-processing definitions” [1].

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The definition of record

- `docs/post-processing-definitions.md` defines every post-processing product the package writes.
- Each definition states the requirement a product is built to.
- Where code and page disagree, the page is right and the code is a defect.

Takeaway *Read the definitions before changing a reduction, an averaging window or a product's columns. The implementation is not its own specification.*

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Part 1

The vocabulary

Turns, passages and windows

Term	Meaning
Revolution	One full turn of a rotor: $60/\text{rpm}$ seconds, $60/(\text{rpm dt})$ solver steps
Blade passage	One revolution divided by the blade count
Azimuthal position	Where a blade is in its turn, 0 to 360 degrees
Window	An inclusive, 1-based range of solver steps that a reduction averages over

Rotor names and reference points

Term	Meaning
SMRP	A rotor's own static moment reference point, <ALIAS>_SMRP
MRP	The global moment reference point of the aircraft
Alias	The name of a rotor, and of the integration group built from its families

Part 2

What every product states

The condition block: angles and velocities

Every table the post stage composes states its condition in one block, in the order shown across these two frames.

Column	Unit	Meaning
ALPHA, BETA	deg	Angles the solver reports it ran at, as the row wrote them
MACH	-	Mach number of that point
RE	millions	Reynolds number the solver reports
VINF	m/s	Free-stream velocity the solver reports
VREF	m/s	Solver reference velocity, used to normalise a coefficient

The condition block: atmosphere and references

Column	Unit	Meaning, continuing the block
ALT	ft	Altitude the row states; NA where it states none
RHO	kg/m ³	Air density the run resolved for that point
TEMP	K	Air temperature the run resolved for that point
MU	Pa s	Dynamic viscosity, written in scientific notation
J	-	Advance ratio the row requested; NA on a row that turns no rotor
J_CLOCK	-	Advance ratio the clock rotor ran at, $V/(nD)$; NA where the record or the reference does not say
RPM_CLOCK	rev/min	Speed the clock rotor turned at, with its hand; NA with no clock
SREF, CREF, BREF	m ² , m, m	Reference area, chord and span

A coefficient needs its normalisation

A coefficient is a force divided by

$$\frac{1}{2}\rho V^2 S.$$

- Area alone does not let a reader recover a force or compare campaigns. Density and the velocity used in the division must be stated too.
- ALT cannot stand in for density: a row may pin density directly.

Which velocity normalises which product

- The solver uses VREF. The steady polar's twenty-four coefficients are the solver's own and use VREF.
- The plots table, every reduction of it and the unsteady polar are rescaled to the free stream by $(VREF/VINF)^2$ and use VINF.
- The rotor table recovers Newtons with VREF, undoing the solver's division. Its coefficients use $\rho n^2 D^4$ and carry no velocity.

Takeaway Both velocities are stated. When they differ, post warns and names the point.

Reference area and chord must match the export

- The solver divides by the area and length in the project it opened. The loads export prints both.
- The package sets neither. It states the reference artifact's SREF and CREF, and checks them against the export.
- A difference beyond the export's printed precision is a **warning** in `post.log` naming both numbers, and every computable product is still written; `--check-frozen` refuses the simulation's products instead.

Takeaway *A reference beside a coefficient must be the reference that divided it.*

Where the condition block appears

- The steady polar already carries ALPHA, BETA, MACH and RE among its twenty-four columns. It adds the rest beside them.
- `probes/<point>_plots.csv` states **no condition**. It keeps the export's own header.
- Reductions read every column of that plots table back as a plotted quantity.

Part 3

The axes of a steady polar

One exported force and moment, three axis systems

The loads export gives one force and one moment per surface in the geometry frame: **x aft, y right, z up**. Sum over the group's surfaces, then turn that pair.

Columns	Axes	How
CDB, CYB, CLB, CRB25, CMB25, CNB25	Body: forward, right, down	Half a turn about y. CDB = C_x and CLB = C_z of the export
CDS ... CNS	Stability	Body axes turned by $-\alpha_s$ about y
CDW ... CNW	Wind	Stability axes turned by β_w about z

Force signs and moment normalisation

- Drag opposes $+x$ and lift opposes $+z$ in each system. Side force keeps its sign.
- Turn the moment as **one vector in one length**, then normalise.
- Divide CR and CN by the span, and CM by the chord.
- Under sideslip, this moves pitch into roll by the ratio of chord to span.

The rotation angles come from the flown velocity

The solver turns sideslip about body z first, then incidence. It flies

$$(u, v, w) = V(\cos a \cos b, \cos a \sin b, \sin a), \quad \alpha_s = \text{atan2}(w, u), \quad \beta_w = \arcsin(v/V).$$

- These angles equal the written ALPHA and BETA whenever either is zero. Otherwise they differ at second order.
- At ALPHA 4, BETA 2, they are 4.0024° and 1.9951° .
- The ALPHA and BETA columns keep what the row wrote.

Wind-axis drag and the solver's integrals

$$CDW = CDi + CDo.$$

- Wind-axis drag from the export's own vector equals the drag the solver integrates, including under sideslip.
- Recorded exports confirm this to their printed precision.
- CDO and CDI state the solver's two integrals.

Wind-axis lift comes from the exported vector

- CLW is the vector's wind-axis lift, not the export's CL.
- On 27 of 28 recorded lifting exports with lift above 0.05, the printed CL is 0.10 to 0.25 per cent higher. One is 0.71 per cent higher. The cause is unknown.
- Every polar axis column uses one source, keeping CDB and CLB consistent with the export's Cx and Cz.
- Older tables used the solver's CL directly, so their CLS and CLW are higher by those amounts.

Recorded evidence: `tests/tier1_offline/fixtures/recorded_total_rows.csv`.

An induced drag the solver did not compute is NA

- A boundary on the vorticity induced-drag list (`SET_VORTICITY_DRAG_BOUNDARIES`) without a defined trailing edge is not computed, and the export prints its `CDi` as zero [2, p.202].
- When the point's run record puts a surface on the list and its printed `CDi` is exactly zero, the group's `CDI` is `NA`, and so is every axis column the export's `x` force reaches at that point's angles: `CDB`, `CDS`, `CDW`; `CLS` and `CLW` off zero incidence; `CYW` under sideslip.
- The moments, `CYB`, `CLB`, `CYS` and `CDO` keep their numbers; the fixed-width custom polar writes `nan`. `post.log` names the declined surfaces.
- A trailing-edged surface whose induced drag rounds to zero at the printed precision reads `NA` too; `significant_digits` narrows that band.

Part 4

The sections table

One export contains every declared distribution

- `sections/<point>_sections.csv` is one solver export.
- It holds every distribution the pproc declares: the wing in XZ, each blade in its own frame.
- The distributions follow one another with no marker between them.
- Leading columns identify which distribution each row belongs to.

The sections table: step and distribution identity

Column	Meaning
STEP	Unsteady: the run's last time step, from the run record. The export is written when the march ends; its own header counts inner iterations, not time steps. Steady: the solver iteration the export states. One name across every table
FAMILY	Geometry families of the row's distribution, joined by +
PLANE	Cutting plane of that distribution

The sections table: rotor identity

Column	Meaning, after PLANE
ROTOR	Rotor whose blades those families are; NA for a surface no rotor owns
AZIMUTH	Where blade one of that rotor is at STEP, in degrees, wrapped to one turn; NA without a rotor

Each rotor has its own azimuth clock

$$\text{AZIMUTH} = \left(\text{blade1.azimuth_deg} + \text{sense STEP} \frac{360}{\text{steps_per_revolution}} \right) \bmod 360.$$

- The datum, the sense of rotation and the steps per revolution all come from **that rotor**.
- Sense is the sign of the rotor's speed.
- Two rotors at two speeds have two azimuths at the same step. A wing has none.

Distribution identity comes from the run record

- The run records which distribution is which. The script states surfaces by index; post cannot recover their names from that alone.
- An older record writes NA in all four identity fields: FAMILY, PLANE, ROTOR, AZIMUTH.
- The same rule applies when the recorded blocks do not add up to the number of exported rows.

The sections table is one instant

- At an unsteady point, the distribution is at STEP. It is not an average over the window.
- Its `products.json` entry states `"kind": "instant"`.
- The history is `series/<point>_sections_series.csv`, with the same identity on every row.

One loads file and one Cp file per distribution

- Each `[[sections.distributions]]` entry gets `sections/<point>_sloads_<name>.csv` and `sections/<point>_cp_<name>.csv` when its export is available.
- The name preserves the alias or joins the declared families with `-`. Filename-invalid characters become `_`; collisions receive the entry's 1-based position suffix.
- All planes and expanded blade blocks of that entry share its file. Cp now has a table, rather than only a manifest listing.
- With per-step exports enabled, every exported step appears in ascending STEP order. Without them, the file holds the end-of-run export.
- The combined sections table and combined sections series remain available.

Distribution columns and recorded identity

- Rows start with STEP, time_s, FAMILY, PLANE, ROTOR, AZIMUTH, then the condition block and export columns. Cp also carries SECTION, the export's 1-based cross-section index.
- Missing values are NA. The manifest records the distribution, original families and steps_tabled.
- A new run records each block's distribution entry. A record can be split only when its recorded layout and pproc identify the blocks unambiguously by families, plane, frame and count.
- Missing or ambiguous identity, mismatched counts, malformed exports and unavailable steps are named skips in products.json. Editing today's pproc cannot supply missing recorded identity.

Part 5

time_average

One average of the whole point

- `time_average` is one average of the whole point over one window.
- An unsteady POLAR row is built from this average.

Takeaway *One window per point, not one per rotor: a point has one history.*

Part 6

per_blade

One shared window, one row per blade

- `probes/<point>_per_blade_<ALIAS>.csv` averages each blade over the last part of the run: the matrix row's averaging window.
- Every blade shares that window. Each row states that blade's start and end azimuth.
- The window is the one the unsteady plots and POLAR use, from `LAST_REVS_AVG` or `LAST_ITERS_AVG` on the matrix row.
- There is one row per blade.

The same time interval makes blades comparable

- Each blade starts at a different azimuth. Its azimuth columns state that difference explicitly.
- A separate passage window for each blade would also sample each blade at a different time, mixing azimuth and history differences.
- Averaging from step one mixes the transient with the answer.

Takeaway *The last shared window keeps the sampling time the same for every blade.*

The leading columns of each blade row

Each row leads with these fields, in this order:

1. REDUCTION, ROTOR, BLADE, FAMILY.
2. The window: FIRST_STEP, LAST_STEP, STEPS.
3. AZIMUTH_START, AZIMUTH_END.
4. The condition block, then the moment point.

Blade plot columns share one heading

- A blade's columns are plots named for its family. A plot is `<parameter>_<group>`; a group cut per blade is named for the blade's family.
- Blade one's `CL_MRP_Blade1` and `FX_LOCAL_Blade1` become `CL_MRP` and `FX_LOCAL` in its row. Removing the family makes the blades line up under one heading.
- The pproc requests these with `families = "each"` or `frame = "LOCAL_AXIS"` over the rotor.

Start and end azimuth belong to that blade

At the first and last step of the shared window, use

$$\text{azimuth} = \left(\text{blade1.azimuth_deg} + \text{blade position} \frac{360}{\text{blades}} + \text{sense step} \frac{360}{\text{steps_per_revolution}} \right) \bmod 360.$$

- `blades` is the rotor's count. A periodic sector carrying two blades of four spaces them a quarter turn apart.
- Without the rotor's clock, the azimuths are NA. The averages are still written.

The reference identifies the blade families

- The rotor block's `families_blades` in the reference artifact identifies its blades.
- The run also records these families.
- A row using flat rotor keys and citing no block gets no per-blade table. `products.json` states why.

Part 7

phase_locked

A mean at a fixed azimuth across revolutions

1. Take the last `last_revolutions_avg` revolutions.
2. At each azimuth, average the samples at that same position across those revolutions.
3. Tabulate the result by azimuthal position, 0 to 360 degrees.

Takeaway *With three revolutions, three datapoints enter each azimuth's mean. The row is an azimuthal position, not a passage, revolution or blade.*

Blade and alias means in one file

- Do the fixed-azimuth average for each blade on its own, about that rotor's SMRP.
- Then do it for each alias, about the global MRP.
- Put all of it in one file with columns suffixed `_{alias or blade_name}_{SMRP or MRP}`.

The azimuthal table's leading columns

`probes/<point>_phase_locked[_<ALIAS>].csv` leads with:

1. REDUCTION, ROTOR, AZIMUTH, STEP, REVOLUTIONS.
2. The steps the revolutions span: FIRST_STEP, LAST_STEP, STEPS.
3. The condition block and the moment point.
4. The plotted columns, under the names the export prints.

Rows use the rotor's last revolution

- There is one row per solver step of the rotor's last revolution, sorted by azimuth from 0 towards 360.
- AZIMUTH is where blade one is, using the sections-table formula.
- STEP is the step of the last revolution that azimuth falls on.
- REVOLUTIONS is the number of samples entering the mean. It equals `last_revolutions_avg` when that is a whole number.

Blade columns align by each blade's own azimuth

- A column ending in a blade family of the rotor is sampled where **that blade** is at the row's azimuth.
- The blade sits its position times $360/\text{blades}$ after blade one. This offset makes two blades line up azimuth for azimuth.
- Every other column, including a rotor total or the aircraft, is tabulated by blade one's azimuth.

Azimuthal suffixes come from the plots

- A plot is `<parameter>_<group>`. The pproc names the group.
- The suffix `_{alias or blade_name}_{SMRP or MRP}` is what a group named `PUSHER_SMRP` or cut per blade prints.
- The package does not rename these columns.

Samples between history steps are read linearly

- One revolution earlier falls a whole number of steps earlier only when `steps_per_revolution` is whole.
- A blade offset falls on a whole step only when the steps per revolution divide by the blade count.
- Where both hold, every sample is a history row and nothing is interpolated. Between steps, the history is read linearly.

The azimuthal mean needs a rotor clock and depth

- Without blade one's datum or the rotor's signed speed, an azimuth cannot be stated. The table is a named skip.
- Declare the rotor in the reference and have the matrix row cite it.
- A depth below one revolution is a named skip too.

The pproc declares the gate and averaging depth

```
[phase_locked]
min_revolutions      = 4.0
last_revolutions_avg = 2.0
```

- `min_revolutions` is the minimum **total** revolutions required for the table to exist. Equal or greater generates it.
- Compare the revolutions the matrix row states it turns, not the length of the exported window.
- `last_revolutions_avg` is the depth of each azimuthal mean and may not exceed `min_revolutions`.

The gate and depth can change at post

- The declared `[phase_locked]` table binds.
- `pyfs-matrix post` reads the current pproc again.
- Editing the gate or depth needs no solver re-run.

A short run loses only the azimuthal reduction

- A row turning fewer total revolutions than the minimum skips this reduction.
- Its POLAR, per-blade table and every other product are written as usual.

Takeaway *Missing the threshold skips the reduction; it does not refuse the run's products.*

An undeclared table keeps the passage series

- A pproc without [phase_locked] still gets the averaging window cut into consecutive blade passages, one row per passage.
- This is the passage series written under this name.
- Declaring the table selects the fixed-azimuth mean instead. Leaving it absent preserves the passage series.

Takeaway *Absent is not zero: there is no minimum gate on the undeclared passage series.*

Part 8

The averaging window

The matrix row states the averaging window

Column	Run type	Required	Unit
LAST_REVS_AVG	unsteady_rotor	Yes	Last revolutions; accepts a float
LAST_ITERS_AVG	unsteady	Yes	Last iterations

The window belongs on the matrix row beside DELTA_TIME, TIME_ITERATIONS and RPM. It does not belong in the pproc: these quantities define its temporal setup.

Window keys use the exact upper-case spelling

- `VAR_NAMES_VALUES` keys match the exact spelling.
- A lower-case `last_revs_avg` on a workflow row is refused as a key of no run type.
- Where that check does not run, the key is not read at all and its intended window is not applied.

Retired window keys are refused; the refusal names the key to write

- `WINDOW_STEPS`, `WINDOW_REVOLUTIONS` and `WINDOW_DEGREES` are retired.
- A row using one is refused at plan time, naming the replacement, even beside its replacement: `LAST_ITERS_AVG` for steps, `LAST_REVS_AVG` for revolutions, degrees divided by 360.
- These keys bind the `time_average` window and the passages cut from it. They do not bind the unsteady `POLAR` window.

New plans and older records without current keys

- A new plan with **no window key** is refused.
- A retired WINDOW_* key no longer permits planning: it is refused with the key to write instead.
- An older record without the current keys is averaged over the window the run was given, with a warning naming the steps.

Retired compatibility forms are refused by name

Each retired form is refused, naming its replacement: `iteration=` (use `step=`), `run.assess_unsteady_from_plots` (use `LoadsAssessor`), the three `WINDOW_*` row keys (`LAST_ITERS_AVG` or `LAST_REVS_AVG`; degrees divided by 360) and group-member lists (one alias per group, members in the reference's `[aliases]`). The `broken_commands` reader stays as plain compatibility with no countdown; existing run records are never rewritten.

Part 9

The unsteady POLAR

The unsteady POLAR averages the plots history

- The source is the plots history, time-averaged over the row's window. It does not read the native coefficient export for that average.
- Columns are plot variables under the names the export prints, rather than the steady polar's fixed twenty-four columns.
- Flight-condition and reference-length columns are added from the workspace.
- The native export states only the last time step, one instant of a rotor cycle. It still ships as a health check.

One file per simulation, one row per point

polars/P<sim>_<name>_uns_avg.csv

- Every file under post/ comes from a sweep. The name states that this is an average of unsteady history.
- It carries the P used by every per-point product.

Each polar row states its window and moment point

1. FIRST_STEP, LAST_STEP, STEPS open the row and identify that point's averaging window.
2. The condition block follows.
3. Then XMOM, YMOM, ZMOM state the moment point.

Takeaway *The plots carry moments. A moment needs its point.*

The super-file content joins the polar

After the plot columns, the table adds:

- The matrix row's cells and the run record's scalars.
- Each rotor's speed.
- The solver flags.

Takeaway *For an unsteady point, this content is part of the polar, not a second super file.*

Each global plot group supplies eighteen axis columns

- The axis coefficients follow the plot columns, using the steady polar's names in this order: CDW .. CNW25, CDS .. CNS25, CDB .. CNB25.
- Each name takes the plot group's **whole** name as a suffix, for example CLW_MRP_TOTAL.
- A second global-frame group adds its own eighteen columns and renames none.

The unsteady axes use dimensional global loads

- Use FX, FY, FZ, MX, MY, MZ from a pproc plot group with frame = "MRP", in Newtons and Newton metres.
- Average these components over the row's window. Divide by that row's $\frac{1}{2} \text{RHO} \text{VIN}^2 \text{SREF}$ and also by CREF for moments.
- Turn them as the steady-polar axes are turned, with the same force signs and final moment normalisation.
- A rotor's own frame cannot supply this block: its axes are not the geometry's axes.

Missing global plots and the axes block

- If the pproc has no global-frame group plotting all six components, the run adds MRP_TOTAL over every boundary, where the run has an MRP frame.
- A pproc that already plots them stays unchanged, to the byte.
- An older run without those plots has no axes block. `products.json` records this under `polars/<file>#axes`.

Takeaway *Missing source plots do not produce an axes block filled with NA.*

The pproc can declare output column names

The package cannot infer which plot label carries which coefficient. Undeclared names pass through exactly as the export prints them.

```
[names]  
CL_MRP_TOTAL = "CL_TOTAL"  
FX_HUB_PUSHER = "FX_PUSHER"
```

This dictionary maps an exported plot column to the name a downstream reader wants. Inventing a source label would otherwise risk an entire column of NA.

The names dictionary changes selected headings last

- It renames the unsteady polar and averaged reductions: `time_average` and the passage series.
- `probes/<point>_plots.csv` keeps the export's names because it is the source table.
- Per-blade and azimuthal tables are not renamed: their columns are no longer the export's own names.
- Axes and `[equations]` read export names. The dictionary applies last, to the heading alone.

The whole dictionary applies, or none of it does

- A source column that no plot of the point prints invalidates the dictionary for that point.
- A destination name already carried by the table does the same.
- Every column then retains the export's name, with a warning and `products.json` entries under `polars/<file>#names` and `probes/<point>#names`.

Takeaway A failed rename never becomes a column of *NA*.

Names that are occupied or refused

- CL is taken. The native export's last-step CL, CDi, CDo, Cx and the rest already appear in the polar's setup content as a health check.
- A plotted column cannot take one of those names. CL_TOTAL is available instead.
- Two dictionary entries giving one name, or a name that is not one word, are refused when the pproc is read.

Part 10

Rotor coefficients

Rotor coefficients describe one rotor

- J , C_T , C_Q , C_P , η , η_{AW} : one table per rotor, every column suffixed with that rotor's alias.
- They are meaningful for one rotor. Summing several rotors mixes different diameters and speeds in the normalisation.

$$J = \frac{V}{|n| D}, \quad C_T = \frac{T}{\rho n^2 D^4}, \quad C_Q = \frac{Q}{\rho n^2 D^5}, \quad C_P = 2\pi C_Q \operatorname{sign}(n), \quad \eta = \frac{J C_T}{C_P}$$

- n is the signed speed in revolutions per **second**: its magnitude sets J , its sign multiplies C_P . These are the classical airscrew coefficients [6, 7].

Tip and helical Mach numbers

$$\Omega = \frac{2\pi \text{ RPM}}{60}, \quad M_{\text{tip}} = \frac{\Omega R}{a}, \quad M_{\text{hel}} = \frac{\sqrt{V^2 + (\Omega R)^2}}{a}$$

- Stated for each rotor an `unsteady_rotor` or `qsteady_rotor` row turns, for the rotor of a steady row that states RPM, and for each actuator disc a row names. R is half the rotor's diameter, or the disc's tip radius; V and a are the point's own resolved free stream and speed of sound. At $V = 0$, $M_{\text{hel}} = M_{\text{tip}}$.
- The rotor table ends with `MTIP_<alias>` and `MHEL_<alias>`; `plan.json` and the run record carry both under `rotor_mach`. A point whose condition does not resolve reads NA in both and keeps its row.
- Both are geometric: V is the free stream alone, without the rotor's induced velocity, which near hover adds to the speed the tip sees. At $M_{\text{hel}} \geq 1$ the tip is sonic, beyond the linearised compressibility of a panel method, and the plan warns once, naming every such point, its rotor and its M_{hel} [1, 10].

An unsteady rotor row uses the shared history window

- A steady point's row comes from the loads export.
- An unsteady point's row averages the plots history over the matrix row's window, shared with the unsteady POLAR and reductions.
- The loads export of an unsteady run is the last time step, one instant of a cycle.

The rotor history is a global group over its own families

- Use the rotor's six components FX, FY, FZ, MX, MY, MZ in global MRP, over exactly its general and blade families.
- Find them through a `[[plots.groups]]` entry with `frame = "MRP"` and those exact families, whatever its name.
- If the pproc plots none, the run adds `ROTOR_<ALIAS>`.
- A group in the rotor's own frame cannot supply this history: its forces turn with the rotor and its moments are already about the hub.
- A group that only shares the alias's name but covers other families is not a source either.

A windowed rotor row never substitutes an instant

- A point whose history does not cover the window, or lacks the required columns, is left out and named in `products.json`.
- The unsteady POLAR applies the same exclusion rule.
- The rotor table's manifest entry states `source` and `window`.

ETAW: the full force vector in wind axes

1. Take the full rotor-frame force vector, $[F_x, F_y, F_z]_{\text{rotor}}$.
2. Carry it to the airframe body frame with the **transpose** of the rotor-to-body rotation.
3. Carry it to wind axes with the **AIAA rotation**, using both alpha and beta.
4. Take its X component, F_{x_W} .

ETAW: a dimensionless efficiency

$$CTW = \frac{F_{x,w}}{\rho n^2 D^4}, \quad ETAW = \frac{JCTW}{CP}.$$

- Wind-axis force replaces thrust, normalised exactly as thrust is. CTW enters where CT enters the efficiency.
- ETAW reduces to ETA when the shaft lies along the stream.
- No wind-axis force supplied means NA.
- Two vector rotations preserve the off-axis force components. The cosine of a scalar angle does not.

A static point has only a real zero advance ratio

- Every rotor coefficient is NA at a static point except J, which is 0.00000.
- The export gives coefficients normalised by the run's dynamic pressure. At $V = 0$, that pressure is zero: the real thrust has already been divided away before the package sees it.
- No rotor coefficient can be recovered from that static export.
- A hover figure of merit cannot be offered either: it needs a force the run does not state.

Part 11

The quasi-steady rotor

A rotor solved steady, and what its products are

- A `qsteady_rotor` row holds the blades still and turns the free stream about the shaft at the rotor's speed; every solve is steady. It is valid only for an **isolated**, **axisymmetric** rotor.
- **Sector** (the row states periodic symmetry): one blade and its copies, one steady solve.
Wheel: every blade meshed; in an axial, uniform inflow it is solved once.
- A wheel at incidence, or in a custom inflow, sees a different flow at each blade position. It is solved once per position and the positions are averaged: `PASSAGE_POSITIONS`: `k`, mandatory, $k \geq 1$, at $\theta_i = i(360^\circ/N)/k$, $i = 0 \dots k - 1$, uniform over one blade passage.
- Such a point's products are those of an unsteady point: the loads at each position (`<point>_qs<i>.txt`), and their average where an unsteady point has its window average.

Takeaway *Uniform positions over one passage give an unbiased mean of a signal that repeats every passage [8]. Two positions settle thrust and torque; the in-plane forces and moments want six or more.*

The validity parameter, carried by every product

- Each blade station's 1P reduced frequency,

$$k = \frac{\Omega c}{2 V_{\text{rel}}},$$

compares the time the flow takes to cross the chord with the period of the once-per-revolution change. Below about 0.05 a quasi-steady answer holds; above 0.1 the lag of the circulation and the shed wake matter [10, 11, 12].

- `plan` shows, per wheel point, the per cent of the span with $k > 0.1$ and k minimum, maximum and mean, and warns when that per cent is above zero.

Where it is written

- The datapoint folder: k minimum, maximum and mean; the span fractions above 0.05 and 0.1; the shares of thrust and torque from stations above 0.1.
- Every post product of the point carries them.
- The sections product carries k at each station.

With `--inflow-fft` a custom inflow's harmonics count too: $k_{\text{eff}} = n_{95} k$, n_{95} counted by one blade per revolution (Guide 1).

Part 12

What the pproc asks the solver
for

Force plots: [plots] parameters

In the pproc	The solver's PARAMETER	Units the package requests	Plots column
CL	CL	COEFFICIENTS	CL_<group>
CDI	CDI	COEFFICIENTS	CDI_<group>
CDO	CDO	COEFFICIENTS	CDO_<group>
CD	CD	COEFFICIENTS	CD_<group>
FX	FORCE_X	NEWTONS	FX_<group>
FY	FORCE_Y	NEWTONS	FY_<group>
FZ	FORCE_Z	NEWTONS	FZ_<group>
MX	MOMENT_X	NEWTONS	MX_<group>
MY	MOMENT_Y	NEWTONS	MY_<group>
MZ	MOMENT_Z	NEWTONS	MZ_<group>

- Write the short name. The pproc refuses FORCE_X. Omitted parameters means all ten. There is no CY, CM or CN plot.
- The solver also offers KILO-NEWTONS, POUND-FORCE and KILOGRAM-FORCE. The package fixes one unit token per parameter. The plots table rescales coefficient columns by $(VREF/VINF)^2$.
- Global MRP loads FX .. MZ feed the unsteady polar and rotor table. With MRP available, the run adds MRP_TOTAL if no MRP group plots all six, and ROTOR_<ALIAS> if none plots all six over that rotor's families.

Sources under src/pyflightstream/: cases/__init__.py, FORCE_PLOT_PARAMETERS, PlotsSpec; commands/unsteady_solver.yaml, UNSTEADY_SOLVER_NEW_FORCE_PLOT; cases/workflows.py, _pproc_plots, _plot_each_rotors_own_history; post/products.py, write_plots_table.

Probes: `[[probes]]` parameters

The pproc and the solver use the same spelling:

CP_FREE	CP_REF	MACH	VELOCITY
VX	VY	VZ	STATIC_PRESSURE_RATIO

- On the unsteady fluid-plot path, each parameter becomes one plot per vertex: `<PARAMETER><n>`, with `n` counting across entries. The probes table un-pivots these columns to one row per probe and step.
- On a steady run, `EXPORT_PROBE_POINTS` takes no variable list. Its columns are the solver's own.
- An entry with empty parameters emits nothing on either run type.

These six boundary-layer parameters are accepted on builds 26.122, 26.123 and 26.124. An unsupported parameter is refused by name and build:

BL_MOMENTUM_THICKNESS	BL_DISPLACEMENT_THICKNESS
BL_TOTAL_THICKNESS	BL_SHAPE_FACTOR
BL_SKIN_FRICTION	BL_TRANSITION_MARKER

Sources under `src/pyflightstream/`: `cases/__init__.py`, `FLUID_PLOT_PARAMETERS`, `ProbesSpec`; `commands/unsteady_solver.yaml`, `UNSTEADY_SOLVER_NEW_FLUID_PLOT`; `commands/probe_points.yaml`, `EXPORT_PROBE_POINTS`; `cases/workflows.py`, `_emit_one_probe_table`; `post/products.py`, `write_unsteady_probes_table`.

The probe source follows the run type

- On an unsteady row, every `[[probes]]` entry uses fluid plots: drawn lines and cited `points_file` profiles alike.
- Each vertex and selected parameter shares one plot counter.
`probes/<point>_probes.csv` holds both forms together, one row per probe and time step, from the plots history.
- A cited profile is read at plan: a count, then `X,Y,Z,TYPE` CSV rows. Types 0 and 1 become fixed fluid-plot vertices with the entry's frame and scale. Invalid profiles are refused before solving.
- The unsteady path no longer imports or exports standard probe points. Its default output list has seven files instead of eight.

Steady probes and older unsteady records

- Steady rows keep the probe-points source for both drawn and cited probes. A nonempty `parameters` list enables the entry; an empty list disables it.
- On a steady row the list does not filter exported variables: the solver exports a fixed set. Plan warns with the entry number and frame.
- On an unsteady row the list selects the sampled fluid-plot variables.
- An older cited profile exported only as an instant has no history. Re-post skips it with its reason in `products.json`, preserving the drawn-line histories that were recorded. A new run supplies the missing history.

What a point exports: [exports]

Key	File suffix	Solver command	Run type
simulation	.fsm	SAVEAS	Both
loads	.txt	EXPORT_SOLVER_ANALYSIS_SPREADSHEET	Both
tecplot	.dat, .vtk	EXPORT_SOLVER_ANALYSIS_VTK; the package writes the .dat	Both
sections	_cp.txt	EXPORT_ALL_SURFACE_SECTIONS	Both
sectional_loads	_sloads.txt	EXPORT_SURFACE_SECTIONAL_LOADS	Both
probes	_probes.txt	EXPORT_PROBE_POINTS	Steady only
plots	_plots.txt	UNSTEADY_SOLVER_EXPORT_PLOTS	Unsteady only
log	_log.txt	EXPORT_LOG	Both

- Keys are true or false. The original kinds default on for their run type. vtk, csv and force_distributions default off; the steady plot kinds default on (next frame).

What a point exports: [exports] (continued)

- `loads` is required to judge the run, and `simulation` is required too: every point leaves its final `.fsm`. Either one stated false is refused.
- `tecplot` exports every surface as VTK and the package writes the `.dat` from it, per panel, in the reference frame.
- `singularity_strength = true`, a top-level pproc key, off by default, adds the nodal `Singularity_strength` of a native auxiliary `*_native_tecplot.dat`, matched to the same step's VTK by geometry [4]. Off, the point makes no native export, its `.dat` states the strength `NOT_CARRIED` (so does its `products.json` entry), and the point is complete without it. `plan` says per row whether the strength is carried.

Sources under `src/pyflightstream/`: `cases/__init__.py`, `EXPORT_KINDS`, `default_outputs`, `_known_export_kinds`; `commands/solver_export.yaml`, `EXPORT_SOLVER_ANALYSIS_VTK`; `results/surface.py`.

Additional exports: plots, a volume section, force distributions

- **The solver's own plots.** `plot_residuals`, `plot_loads` (on by default) and `plot_sections_cp` (on where sections are declared) under `[exports]`: the plotted series as TEXT, one row per iteration or one x/C_p pair per section. **Never a coefficient source:** on 26.124 the plotted pitching moment is -0.897 where the exported CM_y is -0.0247 [3]. An unsteady row saves the first two as well, once, after the march, and the section C_p too, where sections are declared. `false` turns each off.
- **A volume section.** `[volume_section]`: one plane, a rectangle (`corners_m`) or an annulus (`radii_m`), in a plane of a frame, at `offset_m`. **It is sampled, not exported by the solver:** probes on a steady row, fluid plots on an unsteady or rotor row, written as a vertex cloud to `post/<matrix>/fields/<point>_vsec.vtk` or `.dat` (`_step_<STEP>` per unsteady STEP). Older records keep their native export in the datapoint.
- **The force distribution.** `force_distributions = true`: per-panel pressure and viscous force coefficients, once, at the end of the run. Off by default.

VTK and CSV surface exports

`vtk_variables` is a top-level pproc key, before any tables:

```
vtk_variables = ["X", "Y", "Z", "CP_FREESTREAM"]  
[exports]  
vtk = true  
csv = true
```

- Both kinds default off. VTK uses `EXPORT_SOLVER_ANALYSIS_VTK` and `SET_VTK_EXPORT_VARIABLES` when the list is present. An absent list requests all variables.
- CSV uses `EXPORT_SOLVER_ANALYSIS_CSV`: CP-FREESTREAM, PASCALS, all surfaces, solver reference frame. Both exclude the wake.
- Commands and variables are checked against the selected build's database. Unsupported requests are refused at plan, naming the build.

The surface averaging window: the package averages

```
[time_averaging]
last_revs = 1.5 # OR last_iters = 54; exactly one, positive
```

- **The package averages; the solver does not.** The run exports the surface at every step of the window, as EXPORT_UNSTEADY_AFTER_ITER: <first step> would, and the post averages those exports into surfaces/<point>_time_average.dat: the same panel across the steps, every step the same weight, every variable. SOLVER_TIME_AVERAGING is never emitted: it hung 26.124, killed at 240.5 seconds with nothing written [3], and 26.123 does not carry it.

The surface window: time steps, verified on 26.122

- Bounds are inclusive, 1-based time steps ending at the run's last step. `last_iters` counts steps; `last_revs` uses the same rotor clock and rounding as `LAST_REVS_AVG`. A long window clips at step 1.
- Verified on 26.122 [3]: the solver's own average over steps 7 to 12 is the uniform mean of the run's per-step surfaces over those steps, to $1e-13$ in C_p , V_x and speed, which is the package's mean. The solver keeps CF at its last instant; the package averages it.
- Steps that do not share one topology refuse the average; a step of the window that was not exported (a clock-stopped run, a continuation) skips it. Both by name, never a partial average.
- Needs 26.122 or later and `tecplot` on. The export header's inner-iteration count is not this clock.

Surface export provenance and per-step exports

- `products.json` states the averaged surface kind: `average`, with the recorded window, the steps averaged and each per-step VTK read with its sha256; the per-step exports stay kind: `instant`.
- The window is read from the run record (`surface_average_window`), never from a pproc edited since; changing it requires a new run.
- Every `.dat` states its source in `DATASETAUXDATA`: the VTK, its sha256 and a `TRANSLATION` record (cell-centred, reference frame).
- `EXPORT_UNSTEADY_AFTER_REV` or `EXPORT_UNSTEADY_AFTER_ITER` exports the VTK per step, CSV where asked; each step's `.dat` is written from its VTK.
- An older record keeps the solver's window and its meaning. The matrix's plots-reduction window remains a separate choice.
- A storage recipe's `[[prune_step_exports]]` keeps only each point's last step; a post that needs a deleted step refuses, naming it (Guide 1).

Surface formats without a pproc export key

- `EXPORT_SOLVER_ANALYSIS_PLOAD_BDF` still has no `[exports]` key.
`EXPORT_SOLVER_ANALYSIS_FORCE_DISTRIBUTIONS` has one, `force_distributions`.
- Its parameter or payload block cannot be supplied by a setup's one-line `[[raw]]` entry.
- `EXPORT_BL_VELOCITY_PROFILE` holds an unattended script [3]: a raw line is refused at plan on 26.124.

Part 13

The additional post

A second pproc, extracted from the saved simulation

```
ADDITIONAL_PPROC: p002          # VAR_NAMES_VALUES of the row  
pyfs-matrix post matriz.fs --workspace . --additional-pproc
```

- The row runs byte for byte as without the key; the run record never carries it.
- Afterwards, each recorded point whose final .fsm is on disk and hashes as its record says is copied into datapoints/DP-<point>/additional/<pid>/ and reopened there, **with no solve**: sections cut in the frames the run created, sectional loads computed, exports written, file closed. Nothing is saved and no probe is written.
- Each extraction is recorded in additional.json beside runs.json, which is never written. The original .fsm is hashed again after the launch.

Source: docs/post-processing-definitions.md [1], The additional post.

What a reopened saved simulation gives back

Export	Reopened, with no solve (26.124) [3]
total loads	identical
surface solution	identical bytes
surface sections, a distribution created after re-opening included	identical
sectional loads	zero until computed, identical once computed: the file stores the sections, not their loads, so the extraction computes them every time
plots history of an unsteady point	identical
probe points off the body	not identical: up to 4 per cent in speed, 0.094 in C_p

- Only 26.124 was measured: a row on another build is refused at plan.
- Probes, a volume section, [plots], [time_averaging] and base_regions in an additional pproc are refused: nothing measured says a reopened file gives them back.

The products of an extraction

- Under `post/<matrix>/additional/<pid>/`, beside the run's own and never over them: the group polars of a steady point, one sections table per point, and on an unsteady point the plots tables and reductions.
- Every `products.json` entry for them carries:

Key	Meaning
<code>pproc</code>	the ADDITIONAL pproc id, not the one the row ran with
<code>additional</code>	always true; a reader of run products filters on it
<code>extraction</code>	the extractions the file holds, <code><point run id>/additional/<pid></code>
<code>derives_from</code>	the points those extractions were taken from

and no runs, which names run ids everywhere else in the index.

Which rows, which instant, and when it stops counting

- **Rows.** A reopened sections export carries the run's distributions first and the additional pproc's after them. The table keeps every row, the new blocks numbered on after the run's and marked with the pproc; `leading_sections` counts the run's rows.
- **One instant on an unsteady point.** The saved simulation is the LAST instant: the sections table's STEP is the run's last time step, its entry says `kind: instant`, and `post.log` says so. The plots history is the run's own.
- **Only a current extraction has products:** its point is admitted, the `.fsm` still hashes as the one it opened, and every file it wrote is on disk and hashes as recorded. A point run again archives the extraction; the stale one is skipped under `additional/<pid>/runs/<extraction id>`, never under the run's key.

Part 14

What the package does not judge

Last-step convergence does not judge a settled history

- For an unsteady point, the package asserts whether the numerical iterations **within the last time step** converged.
- It does not judge whether the time history has settled.
- There is no settle tolerance or convergence criterion over the history, and no point fails for one.

Takeaway *The user judges whether the history has settled afterwards, from the history.*

Frozen solves and averages

- Two consecutive frozen steps make the point `FAILED_DIVERGED`. In a frozen step, inner iterations after the first print exactly zero velocity residual, and the last prints both residuals exactly zero.
- Assessment names the first frozen step and count. Post reads the native log again, including for a record previously assessed successful.
- An average whose window reaches the first frozen step is skipped with its reason in `products.json`. An earlier stale product is archived.
- Windows wholly before the freeze, raw histories and explicitly instant products remain available. A frozen stretch requires a new solve.

The post never blocks and always writes its log

- Every `pyfs-matrix post` writes `post.log` beside `products.json`, even when nothing is wrong; the manifest names it. Read it after every post: each named skip and warning states the point, the product, the step and what settles it.
- By default a frozen point, an unread residual block, a failed status or a reference mismatch is a WARNING line, and every computable product is written. `--check-frozen` refuses the affected averages instead.
- The freeze and azimuthal guards judge the exact set of plotted samples the reducer reads, blade aliases resolved and every nonzero interpolation weight counted; the guard does no arithmetic of its own.
- `integrate = true` on a `[[sections.distributions]] pproc` entry appends `Strip_length`, `Fx_int`, `Fz_int` and `My_int` to the same `sloads` file: strips midpoint to midpoint, half strips at the ends, the moment about each station's quarter chord. Off by default; the file is byte-identical when off.

One post, one log, and the log as fields

- `post.log.json` sits beside `post.log`: the same header (version, workspace, matrix, time, check_frozen) and one record per WARNING line, in the same order, with point, product, message and remedy. `products.json` names it under `log_json`.
- Both are written from one list, on a clean, a failed and an interrupted post, so they cannot disagree. Parse the JSON, not the text.
- A WARNING line names its own point and product: WARNING
point=camp/sim_7001/AL-020 product=available-exports:
- Each post collects the package's warnings in its own thread: two posts in one process no longer share a log, and a warning from code outside the package is not written to it [3].

Part 15

The token for a value that does not exist

Every product writes NA for no value

- Zero is a real value. Advance ratio at rest is exactly zero, so zero cannot stand for a value that does not exist.
- A blank cannot distinguish missing data, an inapplicable column or a writer that stopped halfway.
- The token is defined once in `pyflightstream._tokens.NOT_APPLICABLE`, below every layer.
- Readers still accept `-`, the token written in older files.

The physics behind the definitions, and where to read it

- **Body, stability and wind axes**, and the angles that turn one into the other, are those of flight mechanics [5].
- **The rotor coefficients** J , C_T , C_Q , C_P and η are the classical airscrew coefficients [6, 7].
- **A window average** is the time mean of a record over a finite span; its error falls as the span covers more periods of the signal [8].
- **The azimuthal (phase-locked) mean** averages the samples that share one phase of the revolution, separating the periodic part of a signal from the rest [9].
- **Tip and helical Mach numbers**, and the reduced frequency that says when a rotor's flow is quasi-steady, are rotor aerodynamics [10, 11, 12].

References

- [1] Geovana Neves, *pyflightstream documentation*, page “Post-processing definitions”.
<https://nevesgeovana.github.io/pyflightstream/post-processing-definitions/>
- [2] Altair Engineering, *FlightStream User Manual*, version 26.12, 2026.
- [3] Geovana Neves, *pyflightstream: measurement records of the solver's behaviour*, 2026.
<https://github.com/nevesgeovana/pyflightstream>, concept DOI 10.5281/zenodo.21482924.
- [4] Geovana Neves, *pyflightstream documentation*, page “Surface exports and native singularity strength”.
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- [5] B. Etkin, L. D. Reid, *Dynamics of Flight: Stability and Control*, 3rd ed., Wiley, New York, 1996.
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